

The Exponent Rules

Exponent Rules Learned in Pre-Algebra

I) $x^a \cdot x^b = x^{a+b}$ Example: $x^2 \cdot x^5 = x^{2+5} = x^7$

II) $\frac{x^a}{x^b} = x^{a-b}$, $x \neq 0$ Example: $\frac{x^9}{x^3} = x^{9-3} = x^6$

New Exponent Rules

III) $(xy)^a = x^a \cdot y^a$ Example: $(2m)^4 = 2^4 \cdot m^4$

IV) $\left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}$, $y \neq 0$ Example: $\left(\frac{n}{3}\right)^7 = \frac{n^7}{3^7}$

V) $(x^a)^b = x^{ab}$ Example: $(x^{10})^3 = x^{10 \cdot 3} = x^{30}$

Strange New Exponent Rules

VI) $x^1 = x$ Example: $3^1 = 3$

VII) $x^0 = 1$, $x \neq 0$ Example: $5^0 = 1$

VIII) $x^{-a} = \frac{1}{x^a}$, $x \neq 0$ Example: $x^{-2} = \frac{1}{x^2}$

IX) $x^{\frac{1}{a}} = \sqrt[a]{x}$ Example: $x^{\frac{1}{3}} = \sqrt[3]{x}$

X) $x^{\frac{b}{a}} = (\sqrt[a]{x})^b$ or $\sqrt[a]{x^b}$ Example: $x^{\frac{5}{7}} = (\sqrt[7]{x})^5$ or $\sqrt[7]{x^5}$

Derivatives of Exponential or Logarithmic Functions and More Practice with Differentiation

Differentiation Formulas

Constant Rule i) $\frac{d}{dx}(c) = 0$ "c" is a constant.

Power Rule ii) $\frac{d}{dx}(x^n) = nx^{n-1}$ "n" is a constant.

Constant Multiple Rule iii) $\frac{d}{dx}[cf(x)] = cf'(x)$ "c" is a constant.

Sum Rule iv) $\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$

Difference Rule v) $\frac{d}{dx}[f(x) - g(x)] = f'(x) - g'(x)$

Product Rule vi) $\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$

Quotient Rule vii) $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$

The Chain Rule viii) If $h(x) = f(g(x))$, then

$$h'(x) = f'(g(x))g'(x)$$

Three Steps For the Chain Rule

- 1) Find the derivative of the outside function while
- 2) leaving the inside function unchanged.
- 3) Multiply the resulting expression by the derivative of the inside function.

Trigonometric Differentiation Formulas

I) $\frac{d}{dx}(\sin x) = \cos x$

II) $\frac{d}{dx}(\cos x) = -\sin x$

III) $\frac{d}{dx}(\csc x) = -\csc x \cot x$

IV) $\frac{d}{dx}(\sec x) = \sec x \tan x$

V) $\frac{d}{dx}(\tan x) = \sec^2 x$

VI) $\frac{d}{dx}(\cot x) = -\csc^2 x$

More Practice Using the Chain Rule with the Quotient Rule

Differentiate.

① $f(x) = \left(\frac{6x^2+5}{x^3+2x-8} \right)^4$

$$f'(x) = 4 \left(\frac{6x^2+5}{x^3+2x-8} \right)^{4-1} \frac{d}{dx} \left(\frac{6x^2+5}{x^3+2x-8} \right)$$

$$f'(x) = 4 \left(\frac{6x^2+5}{x^3+2x-8} \right)^3 \frac{\left[\frac{d}{dx}(6x^2+5) \right] (x^3+2x-8) - (6x^2+5) \frac{d}{dx}(x^3+2x-8)}{(x^3+2x-8)^2}$$

$$f'(x) = 4 \left(\frac{6x^2+5}{x^3+2x-8} \right)^3 \frac{12x(x^3+2x-8) - (6x^2+5)(3x^2+2)}{(x^3+2x-8)^2}$$

$$f'(x) = 4 \frac{(6x^2+5)^3}{(x^3+2x-8)^3} \frac{12x^4+24x^2-96x - (18x^4+12x^2+15x^2+10)}{(x^3+2x-8)^2}$$

$$f'(x) = 4(6x^2+5)^3 \frac{-6x^4-3x^2-96x-10}{(x^3+2x-8)^5}$$

$$f'(x) = \boxed{-4(6x^2+5)^3 \frac{6x^4+3x^2+96x+10}{(x^3+2x-8)^5}}$$

② $y = \left(\frac{x^2+8x}{2x^4-3} \right)^3$

$$y' = 3 \left(\frac{x^2+8x}{2x^4-3} \right)^{3-1} \frac{d}{dx} \left(\frac{x^2+8x}{2x^4-3} \right)$$

$$y' = 3 \left(\frac{x^2+8x}{2x^4-3} \right)^2 \frac{\left[\frac{d}{dx}(x^2+8x) \right] (2x^4-3) - (x^2+8x) \frac{d}{dx}(2x^4-3)}{(2x^4-3)^2}$$

$$y' = 3 \frac{(x^2+8x)^2}{(2x^4-3)^2} \frac{(2x+8)(2x^4-3) - (x^2+8x)(8x^3)}{(2x^4-3)^2}$$

$$y' = 3(x^2+8x)^2 \frac{4x^5-6x+16x^4-24 - (8x^5+64x^4)}{(2x^4-3)^4}$$

$$y' = 3(x^2+8x)^2 \frac{-4x^5-48x^4-6x-24}{(2x^4-3)^4}$$

$$y' = \boxed{-6(x^2+8x)^2 \frac{2x^5+24x^4+3x+12}{(2x^4-3)^4}}$$

$$\textcircled{3} \quad y = \sqrt{\frac{4x-x^3}{3x^5+1}}$$

$$y = \left(\frac{4x-x^3}{3x^5+1} \right)^{\frac{1}{2}}$$

$$y' = \frac{1}{2} \left(\frac{4x-x^3}{3x^5+1} \right)^{\frac{1}{2}-1} \frac{d}{dx} \frac{4x-x^3}{3x^5+1}$$

$$y' = \frac{1}{2} \left(\frac{4x-x^3}{3x^5+1} \right)^{-\frac{1}{2}} \frac{\left[\frac{d}{dx} (4x-x^3) \right] (3x^5+1) - (4x-x^3) \frac{d}{dx} (3x^5+1)}{(3x^5+1)^2}$$

$$y' = \frac{1}{2 \left(\frac{4x-x^3}{3x^5+1} \right)^{\frac{1}{2}}} \frac{(4-3x^2)(3x^5+1) - (4x-x^3)(15x^4)}{(3x^5+1)^2}$$

$$y' = \frac{1}{2 \frac{(4x-x^3)^{\frac{1}{2}}}{(3x^5+1)^{\frac{1}{2}}}} \frac{12x^5+4-9x^7-3x^2-(60x^5-15x^7)}{(3x^5+1)^2}$$

$$y' = \frac{6x^7-48x^5-3x^2+4}{2 \frac{(4x-x^3)^{\frac{1}{2}}}{(3x^5+1)^{\frac{1}{2}}} (3x^5+1)^2}$$

$$y' = \frac{6x^7-48x^5-3x^2+4}{2 \frac{(4x-x^3)^{\frac{1}{2}}}{(3x^5+1)^{\frac{1}{2}}} (3x^5+1)^2}$$

$$y' = \frac{6x^7-48x^5-3x^2+4}{2(4x-x^3)^{\frac{1}{2}}(3x^5+1)^{2-\frac{1}{2}}}$$

$$y' = \frac{6x^7-48x^5-3x^2+4}{2(4x-x^3)^{\frac{1}{2}}(3x^5+1)^{\frac{3}{2}}}$$

$$y' = \boxed{\frac{6x^7-48x^5-3x^2+4}{2\sqrt{4x-x^3}\sqrt{(3x^5+1)^3}}}$$

④ $f(x) = \sqrt[3]{\frac{4x^3-2}{x^4+5x}}$

$$f(x) = \left(\frac{4x^3-2}{x^4+5x}\right)^{\frac{1}{3}}$$

$$f'(x) = \frac{1}{3} \left(\frac{4x^3-2}{x^4+5x}\right)^{\frac{1}{3}-1} \frac{d}{dx} \left(\frac{4x^3-2}{x^4+5x}\right)$$

$$f'(x) = \frac{1}{3} \left(\frac{4x^3-2}{x^4+5x}\right)^{-\frac{2}{3}} \frac{\left[\frac{d}{dx}(4x^3-2)\right](x^4+5x) - (4x^3-2)\frac{d}{dx}(x^4+5x)}{(x^4+5x)^2}$$

$$f'(x) = \frac{1}{3 \left(\frac{4x^3-2}{x^4+5x}\right)^{\frac{2}{3}}} \frac{12x^2(x^4+5x) - (4x^3-2)(4x^3+5)}{(x^4+5x)^2}$$

$$f'(x) = \frac{1}{3 \frac{(4x^3-2)^{\frac{2}{3}}}{(x^4+5x)^{\frac{2}{3}}}} \frac{12x^6+60x^3 - (16x^6+20x^3-8x^3-10)}{(x^4+5x)^2}$$

$$f'(x) = \frac{1}{3 \frac{(4x^3-2)^{\frac{2}{3}}}{(x^4+5x)^{\frac{2}{3}}}} \frac{12x^6+60x^3 - (16x^6+20x^3-8x^3-10)}{(x^4+5x)^2}$$

$$f'(x) = \frac{-4x^6+48x^3+10}{3 \frac{(4x^3-2)^{\frac{2}{3}}}{(x^4+5x)^{\frac{2}{3}}} (x^4+5x)^2}$$

$$f'(x) = \frac{-4x^6+48x^3+10}{3(4x^3-2)^{\frac{2}{3}}(x^4+5x)^{2-\frac{2}{3}}}$$

$$f'(x) = \boxed{\frac{-4x^6+48x^3+10}{3 \sqrt[3]{(4x^3-2)^2} \sqrt[3]{(x^4+5x)^4}}}$$

Finding the Derivative of the Product of Three Things

Differentiate.

⑤ $y = 2x \sin x$

$$y = (2x)(\sin x)$$

$$y' = \left[\frac{d}{dx}(2x) \right] \sin x + 2x \frac{d}{dx}(\sin x)$$

$$y' = 2 \sin x + 2x \cos x$$

⑥ $f(x) = -4x^2 \cos x$

$$f(x) = (-4x^2)(\cos x)$$

$$f'(x) = \left[\frac{d}{dx}(-4x^2) \right] \cos x + (-4x^2) \frac{d}{dx}(\cos x)$$

$$f'(x) = -8x \cos x - 4x^2(-\sin x)$$

$$f'(x) = -8x \cos x + 4x^2 \sin x$$

⑦ $f(x) = -10x^3(x-1)^5 \tan x$

$$f(x) = [-10x^3] [(x-1)^5 \tan x]$$

$$f'(x) = \left\{ \frac{d}{dx}[-10x^3] \right\} (x-1)^5 \tan x + [-10x^3] \frac{d}{dx}[(x-1)^5 \tan x]$$

$$f'(x) = -30x^2(x-1)^5 \tan x - 10x^3 \left\{ \left[\frac{d}{dx}(x-1)^5 \right] \tan x + (x-1)^5 \frac{d}{dx}(\tan x) \right\}$$

$$f'(x) = -30x^2(x-1)^5 \tan x - 10x^3 \left\{ 5(x-1)^4 \frac{d}{dx}(x-1) \tan x + (x-1)^5 \sec^2 x \right\}$$

$$f'(x) = -30x^2(x-1)^5 \tan x - 10x^3 \left\{ 5(x-1)^4 (1) \tan x + (x-1)^5 \sec^2 x \right\}$$

$$f'(x) = -30x^2(x-1)^5 \tan x - 50x^3(x-1)^4 \tan x - 10x^3(x-1)^5 \sec^2 x$$

• ⑧ $y = 4\sqrt{x}(x^2-3)^3 \sec x$

$$y = [4x^{\frac{1}{2}}] [(x^2-3)^3 \sec x]$$

$$y' = \left[\frac{d}{dx} (4x^{\frac{1}{2}}) \right] (x^2-3)^3 \sec x + 4x^{\frac{1}{2}} \left[\frac{d}{dx} [(x^2-3)^3 \sec x] \right]$$

$$y' = 2x^{\frac{1}{2}-1} (x^2-3)^3 \sec x + 4x^{\frac{1}{2}} \left\{ \left[\frac{d}{dx} (x^2-3)^3 \right] \sec x + (x^2-3)^3 \frac{d}{dx} (\sec x) \right\}$$

$$y' = 2x^{-\frac{1}{2}} (x^2-3)^3 \sec x + 4x^{\frac{1}{2}} \left\{ [3(x^2-3)^{2-1} \frac{d}{dx} (x^2-3)] \sec x + (x^2-3)^3 \sec x \tan x \right\}$$

$$y' = \frac{2(x^2-3)^3 \sec x}{\sqrt{x}} + 4x^{\frac{1}{2}} \left\{ 3(x^2-3)^2 (2x) \sec x + (x^2-3)^3 \sec x \tan x \right\}$$

$$y' = \boxed{\frac{2(x^2-3)^3 \sec x}{\sqrt{x}} + 24\sqrt{x^3}(x^2-3)^2 \sec x + 4\sqrt{x}(x^2-3)^3 \sec x \tan x}$$

Simplifying Expressions before Finding the Derivative Differentiate.

• ⑨ $g(x) = x^4(-x^5+2x)$

$$g(x) = -x^9 + 2x^5$$

$$g'(x) = \boxed{-9x^8 + 10x^4}$$

⑩ $h(x) = \frac{10x^4+5x-8}{x^3}$

$$h(x) = \frac{10x^4}{x^3} + \frac{5x}{x^3} - \frac{8}{x^3}$$

$$h(x) = 10x + 5x^{-2} - 8x^{-3}$$

$$h'(x) = 10 - 10x^{-3} + 24x^{-4}$$

$$h'(x) = \boxed{10 - \frac{10}{x^3} + \frac{24}{x^4}}$$

• ⑪ $g(x) = 2x(x^2 - 4x - 6)$

$$g(x) = 2x^3 - 8x^2 - 12x$$

$$g'(x) = \boxed{6x^2 - 16x - 12}$$

⑫ $h(x) = \frac{8x^4 + 2x^3 + 3x^2 - 6}{4x^2}$

$$h(x) = \frac{8x^4}{4x^2} + \frac{2x^3}{4x^2} + \frac{3x^2}{4x^2} - \frac{6}{4x^2}$$

$$h(x) = 2x^2 + \frac{1}{2}x + \frac{3}{4} - \frac{3}{2}x^{-2}$$

$$h'(x) = 4x + \frac{1}{2} + 0 + 3x^{-3}$$

$$h'(x) = \boxed{\frac{1}{2} + 4x + \frac{3}{x^3}}$$

• ⑬ $z(x) = \frac{\sqrt{x} - 1}{x}$

$$z(x) = \frac{x^{\frac{1}{2}} - 1}{x^1}$$

$$z(x) = \frac{x^{\frac{1}{2}}}{x^1} - \frac{1}{x^1}$$

$$z(x) = x^{-\frac{1}{2}} - x^{-1}$$

$$z'(x) = -\frac{1}{2}x^{-\frac{1}{2}-1} + x^{-1-1}$$

$$z'(x) = -\frac{1}{2}x^{-\frac{3}{2}} + x^{-2}$$

$$z'(x) = \boxed{-\frac{1}{2\sqrt{x^3}} + \frac{1}{x^2}}$$

⑭ $r(x) = \sqrt[3]{x^5}(x^3 + 5)$

$$r(x) = x^{\frac{5}{3}}(x^3 + 5)$$

$$r(x) = x^{\frac{14}{3}} + 5x^{\frac{5}{3}}$$

$$r'(x) = \frac{14}{3}x^{\frac{14}{3}-1} + \frac{25}{3}x^{\frac{5}{3}-1}$$

$$r'(x) = \frac{14}{3}x^{\frac{11}{3}} + \frac{25}{3}x^{\frac{2}{3}}$$

$$r'(x) = \boxed{\frac{14}{3}\sqrt[3]{x^{11}} + \frac{25}{3}\sqrt[3]{x^2}}$$

• (15) $z(x) = \frac{\sqrt[5]{x}}{x^2}$

$$z(x) = \frac{x^{\frac{1}{5}}}{x^2}$$

$$z(x) = x^{\frac{1}{5}-2}$$

$$z(x) = x^{-\frac{9}{5}}$$

$$z'(x) = -\frac{9}{5}x^{-\frac{14}{5}}$$

$$z'(x) = -\frac{9}{5x^{\frac{14}{5}}}$$

$$z'(x) = \boxed{-\frac{9}{5\sqrt[5]{x^{14}}}}$$

(16) $v(x) = \sqrt[4]{x}(x+8)$

$$v(x) = x^{\frac{1}{4}}(x+8)$$

$$v(x) = x^{\frac{1}{4}+1} + 8x^{\frac{1}{4}}$$

$$v(x) = x^{\frac{5}{4}} + 8x^{\frac{1}{4}}$$

$$v'(x) = \frac{5}{4}x^{\frac{1}{4}} + 2x^{-\frac{3}{4}}$$

$$v'(x) = \boxed{\frac{5}{4}\sqrt[4]{x} + \frac{2}{\sqrt[4]{x^3}}}$$

Differentiation Formulas for Exponential and Logarithmic Functions

Rule for Exponential Functions: $\frac{d}{dx}(a^x) = a^x \ln a$, "a" is a constant and $a > 0$.

Rule for base "e" Exponential Functions: $\frac{d}{dx}(e^x) = e^x$

Rule for Logarithmic Functions: $\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$, "a" is a constant and $a > 0$.

Rule for the Natural Logarithm Function: $\frac{d}{dx}(\ln x) = \frac{1}{x}$

Find $\frac{dy}{dx}$.

• (17) $y = 3^x$

$$\frac{dy}{dx} = \boxed{3^x \ln 3}$$

(18) $y = 4^x$

$$\frac{dy}{dx} = \boxed{4^x \ln 4}$$

• (19) $y = 7^x$

$$\frac{dy}{dx} = \boxed{7^x \ln 7}$$

(20) $y = 5^x$

$$\frac{dy}{dx} = \boxed{5^x \ln 5}$$

• (21) $y = 6^{x^2}$

$$\frac{dy}{dx} = 6^{x^2} \ln 6 \frac{d}{dx}(x^2)$$

$$\frac{dy}{dx} = 6^{x^2} (\ln 6) (2x)$$

$$\frac{dy}{dx} = \boxed{2x 6^{x^2} \ln 6}$$

(22) $y = 14^{\cot x - 3}$

$$\frac{dy}{dx} = 14^{\cot x - 3} \ln 14 \frac{d}{dx}(\cot x - 3)$$

$$\frac{dy}{dx} = 14^{\cot x - 3} \ln 14 (-\csc^2 x)$$

$$\frac{dy}{dx} = \boxed{-14^{\cot x - 3} (\ln 14) \csc^2 x}$$

• (23) $y = 10^{7x-2}$

$$\frac{dy}{dx} = 10^{7x-2} \ln 10 \frac{d}{dx}(7x-2)$$

$$\frac{dy}{dx} = 10^{7x-2} (\ln 10) (7)$$

$$\frac{dy}{dx} = \boxed{7(10^{7x-2}) \ln 10}$$

(24) $y = 13^{2\cos x}$

$$\frac{dy}{dx} = 13^{2\cos x} \ln 13 \frac{d}{dx}(2\cos x)$$

$$\frac{dy}{dx} = 13^{2\cos x} (\ln 13) (-2\sin x)$$

$$\frac{dy}{dx} = \boxed{-2(13^{2\cos x}) (\ln 13) \sin x}$$

Find $f'(x)$.

• (25) $y = e^x$

$$\frac{dy}{dx} = \boxed{e^x}$$

• (26) $f(x) = e^x$

$$f'(x) = \boxed{e^x}$$

Differentiate.

• (27) $y = e^{\sin x}$

$$y' = e^{\sin x} \frac{d}{dx}(\sin x)$$

$$y' = \boxed{e^{\sin x} \cos x}$$

(28) $f(x) = 2e^{\sqrt{x}}$

$$f(x) = 2e^{x^{\frac{1}{2}}}$$

$$f'(x) = 2 \frac{d}{dx}(e^{x^{\frac{1}{2}}})$$

$$f'(x) = 2 e^{x^{\frac{1}{2}}} \frac{d}{dx}(x^{\frac{1}{2}})$$

$$f'(x) = 2 e^{\sqrt{x}} \frac{1}{2} x^{\frac{1}{2}-1}$$

$$f'(x) = e^{\sqrt{x}} x^{-\frac{1}{2}}$$

$$f'(x) = \frac{e^{\sqrt{x}}}{x^{\frac{1}{2}}}$$

$$f'(x) = \boxed{\frac{e^{\sqrt{x}}}{\sqrt{x}}}$$

• ②⑨ $y = e^{x^2+7x-10}$

$$y' = e^{x^2+7x-10} \frac{d}{dx}(x^2+7x-10)$$

$$y' = e^{x^2+7x-10} (2x+7)$$

$$y' = (2x+7)e^{x^2+7x-10}$$

• ③① $y = \log_2 x$

$$y' = \frac{1}{x \ln 2}$$

③⑩ $f(x) = e^{\frac{1}{x}}$

$$f(x) = e^{x^{-1}}$$

$$f'(x) = e^{x^{-1}} \frac{d}{dx}(x^{-1})$$

$$f'(x) = e^{\frac{1}{x}} (-x^{-2})$$

$$f'(x) = -\frac{e^{\frac{1}{x}}}{x^2}$$

③② $f(x) = \log_5 x$

$$f'(x) = \frac{1}{x \ln 5}$$

• ③③ $y = \log_9 x$

$$y' = \frac{1}{x \ln 9}$$

• ③⑤ $y = \log_3(\csc x)$

$$y' = \frac{1}{(\csc x) \ln 3} \frac{d}{dx}(\csc x)$$

$$y' = \frac{1}{(\csc x) \ln 3} (-\csc x \cot x)$$

$$y' = -\frac{\csc x \cot x}{(\csc x) \ln 3}$$

$$y' = -\frac{\cot x}{\ln 3}$$

③④ $f(x) = \log_{12} x$

$$f'(x) = \frac{1}{x \ln 12}$$

③⑥ $f(x) = \log_{20}(-\frac{2}{x^3})$

$$f(x) = \log_{20}(-2x^{-3})$$

$$f'(x) = \frac{1}{(-2x^{-3}) \ln 20} \frac{d}{dx}(-2x^{-3})$$

$$f'(x) = \frac{1}{-2x^{-3} \ln 20} (6x^{-4})$$

$$f'(x) = \frac{6x^{-4}}{-2x^{-3} \ln 20}$$

$$f'(x) = \frac{-3x^{-1}}{\ln 20}$$

$$f'(x) = -\frac{3}{x \ln 20}$$

• ③⑦ $y = \log_4(-x^2 - 3x + 4)$

$$y' = \frac{1}{(-x^2 - 3x + 4) \ln 4} \frac{d}{dx}(-x^2 - 3x + 4)$$

$$y' = \frac{1}{(-x^2 - 3x + 4) \ln 4} (-2x - 3)$$

$$y' = \boxed{-\frac{2x + 3}{(-x^2 - 3x + 4) \ln 4}}$$

③⑧ $f(x) = \log_8(3\sqrt{x})$

$$f(x) = \log_8(3x^{\frac{1}{2}})$$

$$f'(x) = \frac{1}{(3x^{\frac{1}{2}}) \ln 8} \frac{d}{dx}(3x^{\frac{1}{2}})$$

$$f'(x) = \frac{1}{3x^{\frac{1}{2}} \ln 8} \left(\frac{3}{2} x^{-\frac{1}{2}}\right)$$

$$f'(x) = \frac{1}{3x^{\frac{1}{2}} \ln 8} \left(\frac{3}{2x^{\frac{1}{2}}}\right)$$

$$f'(x) = \frac{3}{6x^{\frac{1}{2}} \ln 8}$$

$$f'(x) = \boxed{\frac{1}{2x \ln 8}}$$

• ③⑨ $f(x) = \ln x$

$$f'(x) = \boxed{\frac{1}{x}}$$

• ④⑩ $y = \ln x$

$$y' = \boxed{\frac{1}{x}}$$

• ④⑪ $f(x) = \ln(\sec x + 2x)$

$$f(x) = \log_e(\sec x + 2x)$$

$$f'(x) = \frac{1}{(\sec x + 2x) \ln e} \frac{d}{dx}(\sec x + 2x)$$

$$f'(x) = \frac{\frac{d}{dx}(\sec x + 2x)}{\sec x + 2x}$$

$$f'(x) = \boxed{\frac{\sec x \tan x + 2}{\sec x + 2x}}$$

Combining differentiation formula xii with the chain rule, we produce the following result.

Rule for Composite Natural Logarithm Functions xiii) $\frac{d}{dx} \{ \ln[g(x)] \} = \frac{g'(x)}{g(x)}$

$$(42) y = \ln(1+x^4)$$

$$y' = \frac{\frac{d}{dx}(1+x^4)}{1+x^4}$$

$$y' = \boxed{\frac{4x^3}{1+x^4}}$$

$$(44) y = \ln(\sin x)$$

$$y' = \frac{\frac{d}{dx}(\sin x)}{\sin x}$$

$$y' = \frac{\cos x}{\sin x}$$

$$y' = \boxed{\cot x}$$

$$(43) f(x) = \ln(7x)$$

$$f'(x) = \frac{\frac{d}{dx}(7x)}{7x}$$

$$f'(x) = \frac{7}{7x}$$

$$f'(x) = \boxed{\frac{1}{x}}$$

$$\bullet (45) f(x) = \ln(x^3 + \tan x)$$

$$f'(x) = \frac{\frac{d}{dx}(x^3 + \tan x)}{x^3 + \tan x}$$

$$f'(x) = \boxed{\frac{3x^2 + \sec^2 x}{x^3 + \tan x}}$$

$$(46) f(x) = \ln(-7x^4 + x)$$

$$f'(x) = \frac{\frac{d}{dx}(-7x^4 + x)}{-7x^4 + x}$$

$$f'(x) = \boxed{\frac{-28x^3 + 1}{-7x^4 + x}}$$

$$(48) f(x) = \ln(\cos x)$$

$$f'(x) = \frac{\frac{d}{dx}(\cos x)}{\cos x}$$

$$f'(x) = \frac{-\sin x}{\cos x}$$

$$f'(x) = \boxed{-\tan x}$$

$$(47) y = \ln(5x^2)$$

$$y' = \frac{\frac{d}{dx}(5x^2)}{5x^2}$$

$$y' = \frac{10x}{5x^2}$$

$$y' = \boxed{\frac{2}{x}}$$

$$(49) y = \ln(9-x)$$

$$y' = \frac{\frac{d}{dx}(9-x)}{9-x}$$

$$y' = \boxed{\frac{-1}{9-x}}$$